RAMA CHIRANJEEVI

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PROFESSIONAL SUMMARY

Automotive R&D; professional with 9 years of experience across CAE simulation, durability analysis, product development and engineering automation, complemented by hands-on work in machine learning, Generative AI, MLOps and digital twin technologies. Experienced across steering, chassis, transmission and sealing systems, with a focus on applying data and AI techniques to engineering workflows and product development. Combines engineering depth with business analytics and strategic problem-solving to support automotive AI and digital transformation initiatives.

CORE SKILLS

Automotive Engineering: CAE, FEA, durability analysis, strength analysis, product development

AI / Machine Learning: Python, NLP, computer vision, deep learning, regression, classification, time series, survival analysis

Generative AI: LLMs, RAG, LLaMA, BERT, LlamaIndex, MCP-based knowledge retrieval

MLOps & Deployment: Docker, FastAPI, Streamlit, AWS EC2, model deployment

Data & Analytics: SQL, statistics, probability, hypothesis testing, A/B testing, data storytelling

Engineering & Data Tools: Abaqus, Ansys, HyperMesh, SimLab, TensorFlow, PyTorch, Git

PROFESSIONAL EXPERIENCE

Technical Specialist | ZF TCI | Hyderabad | Oct 2024 – Aug 2026

- Contributed to simulation and AI-driven product development initiatives within Engineering R&D;, collaborating with cross-functional teams on engineering workflows.
- Performed strength and fatigue analysis for steering components and chassis systems, including Control Arm, Strut and Damper components.
- Supported chassis analysis across strength and fatigue load cases, contributing to a reported 10% reduction in project hours.
- Worked on steering portfolio analysis, including component strength and fatigue assessment.

CAE Analyst | Mercedes-Benz R&D; Centre India | Bengaluru | Jul 2021 – Sep 2024

- Conducted multivariate regression analysis of vehicle mileage performance across parameters including vehicle weight and engine type, informing business decisions and delivering reported cost savings of 15%.
- Architected a German-language learning application for professionals, covering data collection, storage solutions, data architecture and data security frameworks.
- Performed exploratory data analysis and customer segmentation across 40 automotive companies using financial indicators to assess market capitalization and competitive positioning.

Systems Engineer | Tata Consultancy Services | Bengaluru | Oct 2019 – Jul 2021

- Performed structural analysis of automotive sealing components, including valve-body sealing, O-rings and gasket modelling.
- Executed a Six Sigma study on modelling approaches, resulting in a reported 30% reduction in manual modelling work hours.

Senior Engineer | General Motors Technical Centre India | Bengaluru | Aug 2017 – Oct 2019

- Delivered CAE analysis on transmission subsystems, including clutch and dam systems, improving process efficiency by a reported 20% through automation.
- Conducted durability and mount-load analysis for AWD and 2WD platforms, reducing delivery time by a reported 20%.

SELECTED PROJECTS

AI-Powered FEA Convergence Troubleshooting Tool | LLM & MCP

- Built an LLM-driven diagnostic workflow integrating Hugging Face models with MCP-based knowledge retrieval to analyze Abaqus simulation failures.
- Developed a Streamlit interface to parse .inp, .msg and .sta files for error detection and root-cause analysis, generating context-aware recommendations from domain-specific simulation knowledge.

Digital Twin for Automotive Systems | Knowledge Graph

- Designed a knowledge graph-based digital twin model connecting mechanical, electrical and software systems, including engine, ECU, battery and sensors.
- Mapped cross-domain relationships to explore system behavior, dependencies, failure prediction and performance optimization.

End-to-End MLOps Pipeline | Docker & AWS EC2

- Built a customer-conversion prediction pipeline covering exploratory analysis, model training and evaluation; containerized the model with Docker and deployed APIs using FastAPI on AWS EC2.

Enterprise Data Layer Strategy | Data Architecture

- Developed an enterprise data architecture strategy across structured, transactional, unstructured and analytical data layers, evaluating Oracle, CockroachDB, MongoDB, Amazon S3, Snowflake and Amazon Redshift.

Customer Churn & Survival Analysis | Statistical Modeling

- Built churn models using Logistic Regression and Discriminant Analysis; applied Kaplan–Meier and Cox proportional-hazards analysis to derive retention and customer-lifetime-value insights.

Sentiment Intelligence using Fine-Tuned BERT | NLP

- Designed a five-level customer-review sentiment classifier, fine-tuned on 10,000 labeled reviews, and compared pre-trained and fine-tuned model performance using precision, recall and F1-score.

NIFTY 50 Time-Series Forecasting | ARIMA & Smoothing

- Analyzed five years of daily NIFTY 50 closing prices; benchmarked moving average, exponential smoothing, Holt's linear and ARIMA models. The project reported ARIMA as best-performing (MAPE: 0.442).

Strategic Decision-Making & Operations Research

- Analyzed the Merton Truck Company case using linear programming and sensitivity analysis; formulated a set-covering and assignment model to optimize logistics hub locations and demand-point coverage.

EDUCATION

Advanced Management Programme in Business Analytics (AMPBA) — Indian School of Business (ISB), Hyderabad | Completed Apr 2026

M.Tech, Energy Engineering — National Institute of Technology (NIT) Tiruchirappalli | Jun 2015 – Apr 2017 | CGPA: 9.3/10; Top 1% of class

B.Tech, Mechanical Engineering — G. Pulla Reddy Engineering College, Kurnool | Jun 2010 – Apr 2014 | CGPA: 8.83/10; Gold Medalist

INTERNSHIP

Industrial IoT & Predictive Maintenance Intern | KLVIN Technology Labs Pvt. Ltd., Hyderabad | Mar 2026 – May 2026

- Analyzed multivariate motor-sensor time-series data under varying loads, identifying degradation patterns and estimating Remaining Useful Life (RUL) to support maintenance insights.

CERTIFICATIONS

- Microsoft Azure AI Essentials Professional Certificate — Microsoft, 2026
- Introduction to Digital Twin — LinkedIn Learning, 2026
- Introduction to Generative AI — Google, 2024
- GGI MBA Scholar — GGI, 2022
- Electric Vehicle Development — ARAI, 2021
- Artificial Intelligence Foundations — LinkedIn Learning, 2020
- Lean Six Sigma Green Belt — General Motors, 2017

ACHIEVEMENTS

- Bronze Star, Department Level, Mercedes-Benz R&D; Centre India, for digitalization of eMotor systems using automation techniques.
- B.Tech Gold Medalist, JNTU Anantapur, 2014, for securing the overall college top rank.
- Merit Student recognition, AVOPA Educational Society, 2013.